Progress Report

Weekly Project Update Overview

Project Development Timeline

Weeks 1-3 Sensing Architecture

Determine what sensing architecture to use as well as what vision and measurement to use on the robot.

Weeks 4–6 Mechanical & Actuation

Structural design of the 4-DOF robotic arm, choice of actuators, and the pneumatic gripper.

Weeks 7-9 Control & Programming

Build the control system with ROS2, arduino motor controller and motion planning.

Weeks 10–12 Integrate all subsystems

Connect all subsystems and test the system on the smart pick-and-place operation.

Weekly Progress Goals

DONE

ONGOING

01

ACTUATOR SELECTION

DC motors with encoders (12V) have been considered and chosen as primary actuators of the robotic arm.

02

PNEUMATIC GRIPPER SYSTEM

The pneumatic component evaluation is underway and this will involve the selection of a compressor, control of the pressure, and the budgeting of the necessary hardware.

03

SENSOR SELECTION

The sensing parts are being evaluated, and these components are the vision camera that detects objects, and the force sensors that sense the gripper pressure.

Weekly Timeline

1. Brushless DC motors for the robotic arm were selected.
2. Evaluation of pneumatic components and pressure requirements began.
3. Vision camera and force sensing options were reviewed.
4. Initial budgeting and equipment procurement planning started.

Goal # 1

Actuation System Development

Completed

TRACKING SHEET

PROGRESS CHECK

- Brushed and Brushless DC motors for the robotic arm were evaluated and selected.
- Budget approved for boost/sponsored posts across platforms
- Initial actuator configuration for the robotic arm was defined (4-DOF).

BLOCKERS

- Final motor driver compatibility with the control system still needs verification.

ACTION ITEMS

- Select appropriate motor drivers for the chosen actuators.
- Begin integration planning between motors and the Micro-controller.

Goal # 2

Pneumatic Gripper System

In progress

TRACKING SHEET

PROGRESS CHECK

- Initial research on pneumatic gripper mechanisms completed.
- Compressor and pressure control requirements reviewed.
- Force sensing approach using FSR considered for grip monitoring.

BLOCKERS

- Final selection of pneumatic components depends on budget and hardware availability.

ACTION ITEMS

- Finalize pneumatic actuator and compressor selection.
- Estimate required pressure range for gripping tasks.
- Prepare component list for procurement.

Goal # 3

Vision & Sensor System

In progress

TRACKING SHEET

PROGRESS CHECK

- Vision sensing approach for object detection was reviewed.
- Camera options for the system were evaluated (intel realsense d415).
- Force sensing for gripper feedback was considered.

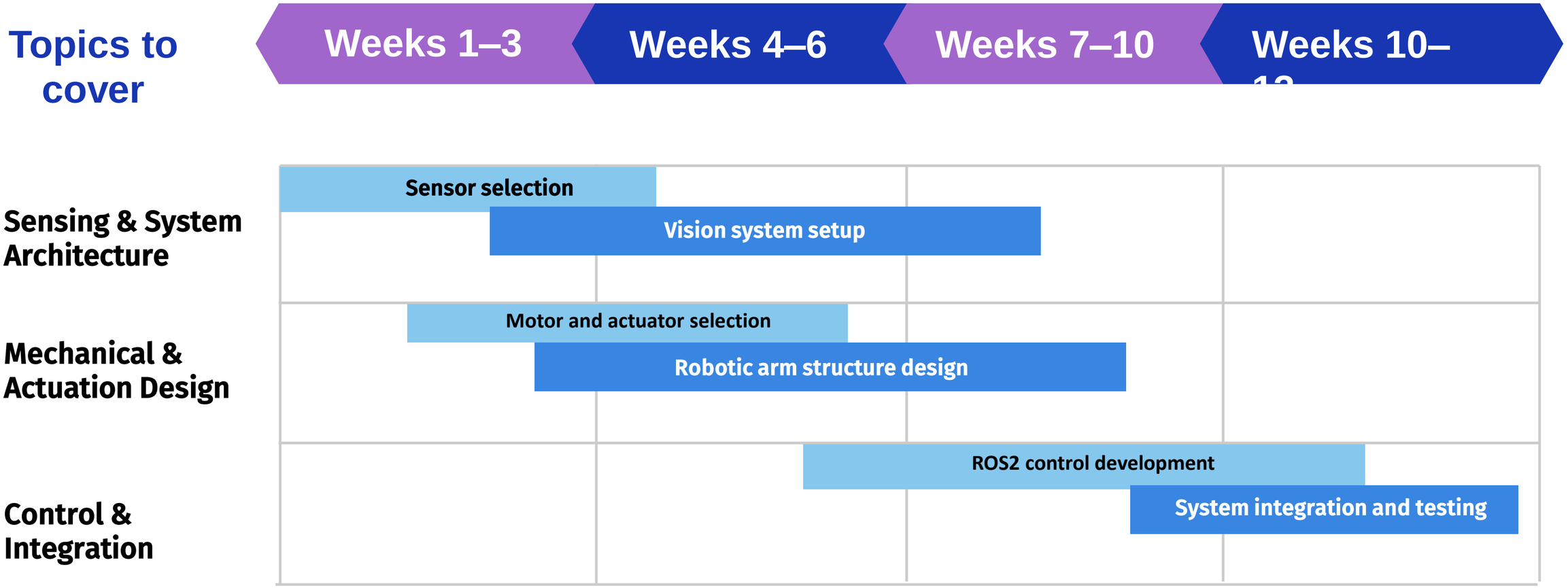
BLOCKERS

- sensor fusion depends on integration with control system which is still underprocess.

ACTION ITEMS

- Finalize vision sensor selection.
- Define sensing method for object detection (OpenCV color based and object based).
- Prepare sensor integration plan.

Project schedule (Gantt)



Omar’s progress

Time frame:

Weeks 1-3

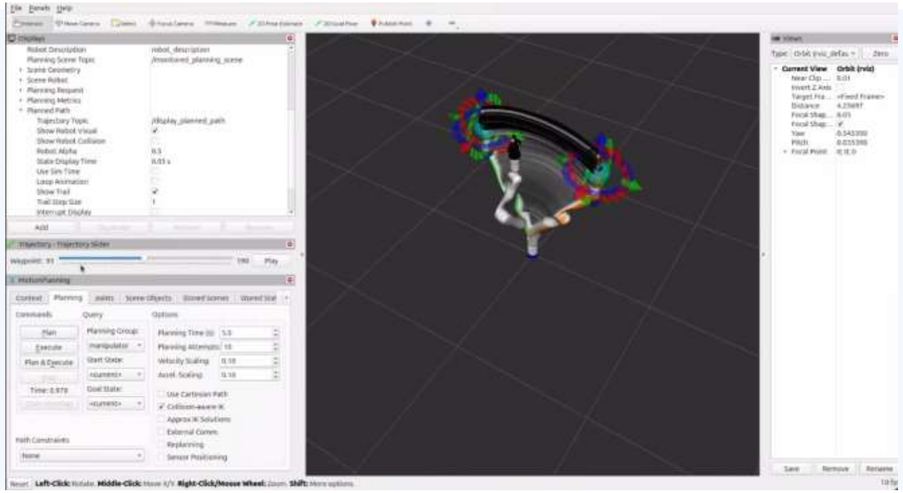
Main role:

Software & Ros2

Accomplished (since last report)

- Defining what the robot needs in terms of computer vision and object detection
- Started using OpenCV and made comparisons with YOLO for classifiers
- Made a github Repo for the project to better share the work with the group members

Evidence (show, don’t tell)



Next Week Plan

- Use a model for object detection and tracking of a red ball and connect a depth camera and see how it behaves

In Progress

- Implementing Moveit on an already made 6-DOF robotic arm and familiarizing myself with the process

Help Needed / Questions

- Hardware specs confirmation
- Motor driver compatibility
- ROS2 integration issues

Hamza’s progress

Time frame:

Weeks 1-3

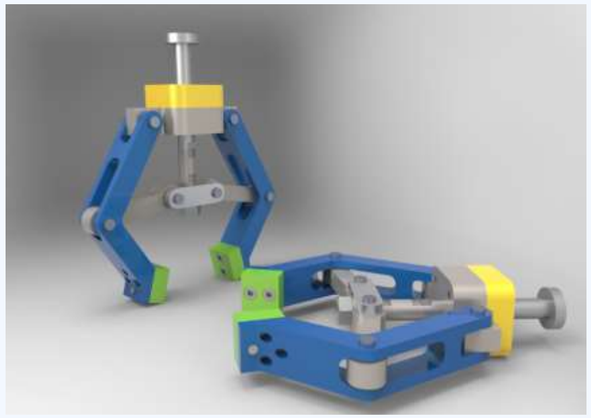
Main role:

Gripper(pneumatic)

Accomplished (since last report)

•Carried out the first study regarding pneumatic gripper mechanisms that could be used in pick-and-place robotic arms.
•Researched on various pneumatic actuation techniques such as single-acting and double-acting cylinders in gripping motion.
•Examined important aspects of pneumatic component.

Evidence (show, don’t tell)



Next Week Plan

•Complete pneumatic cylinder and gripper.
•Make a detailed list of pneumatic parts to be bought.
•Begin CAD design of the gripper mechanism and mounting interface with the robotic arm.
•Evaluate the integration of solenoid valve control with the microcontroller.

In Progress

•compare the various types of pneumatic cylinders and gripper designs in order to come up with the best suited one to use in the robotic arm..
•To determine the range of pressure and amount of air needed to be dependable in gripping.
•The development of a simple gripper framework which can fit the robotic arm end-effector.

Help Needed / Questions

Confirm compressor pressure and specifications.
Instruction on how to set the solenoid valve to the microcontroller..
Verify compatibility with the robotic arm mounting interface.

Khaled's progress

Time frame:

Weeks 1-3

Main role:

Actuator selection

Accomplished (since last report)

- Discussed actuators that can be used in robotic arm systems with the focus on servo (or DC), brushed and brushless motors.
- In comparison to three of the three actuators founded on torque, speed, precision of movement, and the ability to interface with Raspberry Pi, Arduino systems, and ROS2 systems.
- Started screening types of magnetic sensor that would be able to distinguish or detect nuts and screws.

Evidence (show, don't tell)

1) Actuator decision matrix (joint / platform motion)

Options

- A) **Servo motor system** (motor + feedback sensor + servo drive)
- B) **Stepper motor** (open-loop or with encoder)
- C) **Brushed DC motor + encoder** (closed-loop with controller)

Criteria	Weight	Servo Motor	Stepper Motor	BLDC Motor	Brushed DC Motor + Encoder
Torque Capability	0.25	4	3	4	4
Speed Range	0.20	3	2	5	4
Position Accuracy	0.25	5	4	3	4
Control with ROS2 + Arduino	0.15	4	4	3	5
Cost & Availability	0.15	3	4	2	5
Weighted Score	1.00	3.90	3.30	3.45	4.30 (Winner)

Next Week Plan

- Choose the types of actuator used and then finalize the specifications for their attached motors for the robotic arm before continuing through this step.
- Determine the appropriate motor drivers that are compatible with your preferred motor type and selected micro controller(s).
- Verify the configuration of magnetic sensors and how they can be utilized for detection of metal objects to classify them.

In Progress

- Considering the torque and speed criteria of the robotic arm joints.
- Research on combination of actuators with ROS2 control system.

Help Needed / Questions

- To check the ultimate hardware requirements of the robotic arm.
- Find out the motor drivers that would be used with the chosen actuators..
- Recommendations on how to integrate and program the actuator and motor with a robotic arm based on ROS2.

Abdelaziz’s progress

Time frame:

Weeks 1-3

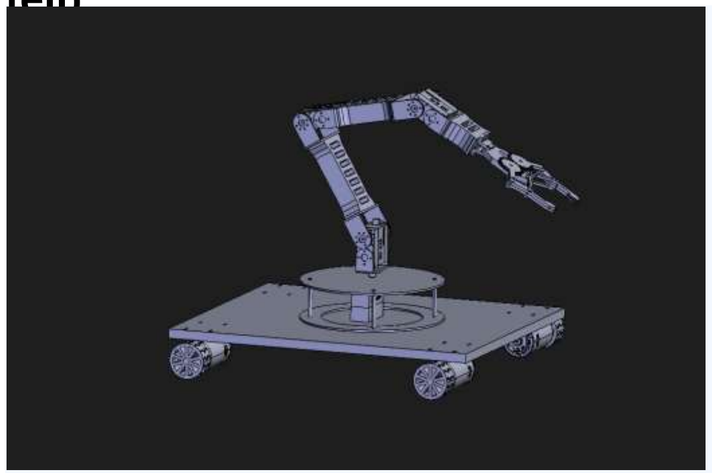
Main role:

Mechanical structure

Accomplished (since last report)

- Researched various designs of robotic arm and industrial arm.
- Choose number of Degrees of Freedom (DOF) and joint movement.
- Choosing a 6-DOF robotic arm concept.
- Created the initial mechanical design (base, links, joints).
- Researched the materials and dimensions to be used..

Evidence (show, don’t tell)



initial conceptual Design.

Next Week Plan

•Fully realistic model of the robotic arm CAD.
•Complete the size and the connection lengths of the building.
• Choose appropriate motors or actuator of each joint.
•Determine the design of the mounting of the robotic arm on the mobile platform.
•Begin basic loading tests or mechanical simulation of the design.

In Progress

- Optimizing the mechanical design of the arm links and joints.
- Assessment of various joint mechanisms in order to move smoothly.
- Completion of CAD assembly of complete robotic arm system.

Help Needed / Questions

- Recommendations on the choice of appropriate motors/servos to be used at each joint.
- Suggestions on material of light yet stiff robotic arm links.
- Assurance of the best number of DOF required in the project.
- Response to the present CAD plan of the arm design.